

Computer Commands

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Introduction

The design goal is to select elements that will fit on a typical home computer wide screen monitor. There aren't enough square inches to duplicate the multiple displays of the real system. The "extra" room on the side was just about right to hold some old style controls that are still applicable.

You communicate with the ATC Computer via an input/output text window in the style of the CRD, Computer Readout Device, of NAS Stage A. This is the window with the black background at the middle right of the screen. There is only one CRD and therefore no distinction between R-side and D-side CRD functions. Some D-side activity will generate CRD input/output when you are using the radio input/output windows. The only pronunciation I have ever heard for CRD is "crud".

If you start typing with any key except <enter>, the characters go to the CRD input area. Starting with the <enter> key is used to activate the radio to transmit to an airplane. (A white square indicates the transmitter is active.) If CRD input is in progress, <enter> will terminate most input and start the processing. A mouse click on a data block position symbol (diamond or triangle), or on a present position slash, may be used to end some commands. The Range-Bearing function is different, and ends with a mouse click on a position of interest.

The <esc> or <escape> key is used to cancel and clear input from the CRD (or the transmitter). I use <esc> a lot to correct mistakes. Some unusual key sequences require a second <esc> to clear the input.

Most CRD inputs produce a response beginning with ACCEPT or REJECT in the output window. The data is also written to the log file. It seems better to *not* echo data block orientation changes because the result of the previous action is probably more useful. However, an existing REJECT message will be cleared to prevent confusion.

The input window will be cleared after an ACCEPT message. Most REJECT messages leave the input so you can edit it using standard Windows operations (left arrow, delete, etc.), then hit <enter> again.

The buttons in the lower right corner with the light blue background also make inputs to the computer in the style of QAK (Quick Action Keys) and rotary switches of the 1980s, plus a few simulation-specific items. Buttons can be clicked with the mouse, and should work if you have a touch sensitive screen. Some of the elements in this panel have keyboard equivalents. [add link] See table at the end of this section. ERAM equipment has similar functions but different details. It is not terribly difficult to modify labels and layout (keeping screen space constraints in mind) to make it more up to date.

Standard Windows keyboard function keys may be used for QAKs if shown in white text below the button. If the radio transmitter is not active, these keyboard function keys simply enter characters into the CRD input window to get a command started (e.g. QP<space>). Most of the QAKs also display a hint in the output window of an expected input format. There may be additional legal formats as well.

One option for the RNG/BRG (range/bearing) key is no text input, just two left clicks on the points of interest. Another option is RNG/BRG <Fix> <left click>, which gives the great circle range/bearing TO the named fix FROM the point clicked. If the first click of the former, or the only click of the latter, is on a data block position symbol, the ETA for a straight line path to the fix at the current displayed ground speed is also shown, except for very low ground speeds. The associated line on the display will disappear after 10 seconds (no early cancel), or change if you do another range/bearing.

Additional commands to control the simulation, such as selecting a sector, begin with XX and are also entered into the CRD input area. Early versions of the program only had XX commands, and some now

have menu selections or buttons that are easier to use. Some functions in the real system use mouse actions only, and do not have keyboard equivalents. Until pop-up windows can be programmed, it may make sense to use XX-style commands to perform these functions but give them a more descriptive name, such as CRR to handle Continuous Range Readout mouse actions.

A few XX commands have many parameters. If the desired input is available from another source, a normal Windows copy/paste into the CRD input area will work.

If keyboard or mouse clicks do not work as expected, it is likely that input is being sent to a non-obvious place.

- A window other than the main program window may be active, which can be fixed by clicking within the main window.
- If the mouse cursor is over a Zoom button, Windows considers that button to be active and prevents keyboard input from going anywhere useful. Move the mouse cursor off the button. Clicking on a zoom button will move the cursor off the button automatically.
- If CRD input is in progress or the previous input has been retained due to an error, function keys and unrelated mouse clicks are ignored, and <enter> will not activate the transmitter. Check for a previous REJECT message and clear with <esc>.

Notes on Command Table

The following is terminology used by the program, and may be different from, or not present in, the real system. Some convenience functions may change if there is a similar real world function that can be duplicated.

<FLID> = Flight ID input as text. For most commands, may use Aircraft ID, computer ID (CID), or discrete transponder code. Aircraft ID must be full call sign. (Abbreviated call sign is a radio option.)

<FLID ♦> = above plus left click on position symbol ♦ or ▲. A required right click will be stated explicitly.

Details of one command (interim altitude) from the table below: QQ ddd <FLID ♦>. "ddd" means 3 decimal digits, say 350, meaning flight level 350. If the call sign is N1234X with CID 456, and transponder code 5361, you could use any of:

QQ 350 N1234X<enter>

QQ 350 456<enter>

QQ 350 5361<enter>

QQ 350(plus left click on ♦ or ▲)

QQ<space> may be typed, or press <F8>, or click on the INT ALT button.

<Sector ID> for handoffs. For a sector within the active facility - two digits. For a sector in another facility, one letter plus two characters, or just the letter designating the facility (the computer knows ...). Examples:
23 L35 S1A S

Some often used commands, such as accepting a handoff and repositioning a data block, use the "none" key, which means a QAK (or two letter command) is *not* needed. If extra typing makes you feel good you may start these commands with QN (or QZ or <F5>).

Many functions require that you are plugged in. When you are not plugged in you can do things that do not change track or flight plan data, such as displaying flight plan text or a route line.

If you click on a track symbol that is handing off to the active sector and the data block disappears, it means you are not plugged in. This possibly surprising result is consistent with how you get rid of any data block that you do not own (and in this case can not own because you are not plugged in). If no other action is taken, the data block will appear when the active sector AI accepts the handoff.

A "not plugged in" response to an input that should be allowed when not plugged in may be a typing error. For example repositioning a data block uses one digit for the direction. If you type two digits, it is interpreted as a sector ID and a handoff attempt (which fails if you are not plugged in).

In general, track and flight data commands require that you own the data block. "/OK" is implemented in a few commands to override the normal restrictions. (More can be added.) A proper use of /OK is to operate on a data block within your airspace that is owned by another sector, e.g. unintended handoff. It may also be needed when a program bug doesn't produce a handoff to you. To allow recovery from any type of bug, the program allows gaining control of any datablock, including after a handoff to another facility.

"/TT" can be used to change an interim altitude when you no longer own a data block without stealing ownership. [ERAM?] Real world use would depend on facility procedures and/or coordination. At present, these overrides are applied even if the new owning facility is a Tracon.

If present, the world's fastest D-side will take care of Flight Data items QQ, QS, QU/Track Reroute, AM RAL, AM RTE, and VCI On/Off (if allowed by a menu item), when you transmit to an airplane, and QR when a non-Mode C aircraft reports its altitude to you. The D-side uses overrides if necessary.

A function not related to the D-side is automatic handoff initiation from the active sector. This is a normally enabled computer function that will happen whenever the caret symbol '^' is not shown above or to the left of the data block. More details in eATS.PDF, Automatic Handoff section.

QU/Track Reroute attempts to predict the appropriate sector for handoffs. It may not produce what is desired during altitude changes with shelves, or if the aircraft position at the time of the reroute is outside of the active sector. You should be able to restore a desired handoff target using QZ with an altitude in the target sector. The default track line display shows the handoff sectors because the program isn't going to always get it right. You can turn off that data element with a menu selection if you don't like it.

A switch on the A.I. menu allows or disables the D-side doing a QU to a nearby fix on a SID or STAR. This action can cause problems with the altitude logic and handoff sectors. The default is unchecked, no action by the D-side. The assumption is you have done some minor vectoring and then issue a direct to get back onto the SID/STAR at a fix within your sector, or you gave a small short cut. If auto hand became inhibited due to distance off course, it will re-enable approaching the original course. You can always make the QU entry if you want it.

Command Table

The final element in a command is <enter> (usually not shown), or a mouse click.

A space is required to separate other elements. Using a QAK, function key, or mouse click will automatically add a space before, or a space after.

For Q-commands where '/OK' is implemented, the order must be: Q-item /OK other fields. A low priority item is to allow it anywhere on the input line.

Action	Requirements	Command	Notes
----- TRACK ----- (Most require plugged in)			
Accept Handoff	Data block handing to your sector.	<FLID ♦>	If not plugged in, or is handing to a different sector, this is processed as a display action (which will make the data block disappear).
Accept Override Handoff	Target tracked. Data block has to be displayed if using mouse click.	/OK <FLID ♦>	CAREFUL !!! Handing off later may require a QU/Track Reroute to get things back in sync. Communications not changed (may have to use interphone). If a normal accept would work, /OK is ignored.
Initiate Handoff	You have track control. Target sector is one of the next two in the ATC flight plan. Data block is not already in handoff.	<Sector ID> <FLID ♦>	Will eventually start automatically if auto hand is enabled for this aircraft. You may prefer to initiate earlier as soon as an aircraft is clear of other traffic. Also for non-radar airplanes in CST to make freq changes and other functions work. Additional programming required to make this realistic.
Retract Handoff	Handoff has been initiated on a data block you own and not yet accepted by another sector.	<FLID ♦>	Auto hand is disabled. Manually initiate handoff later and/or enable auto hand (QA) for this aircraft. Field E handoff indicator not updated (cleared).
Auto Hand	If trying to enable, the logic may inhibit again for various reasons.	QA <FLID ♦>	Enables/Inhibits auto hand for one aircraft. Inhibiting to a Sector or a Facility not programmed yet.
Track	Owned datablock in CST.	QT<left click><FLID> QT //<fix> <FLID>	Initial support for non-radar operations. Move datablock in CST to another position. FLID must be typed. No space if after a left click.
Drop Track	Aircraft landed and is in CST.	QX <FLID ♦>	Deletes data block from display. To make internal bookkeeping easier, only works after receiving "landed" report. Before QX is allowed in general, need to program the start track function to recover from mistakes.
----- FLIGHT DATA ----- (Most require plugged in, own data block)			
Assigned Altitude		QZ ddd <FLID ♦>	Change long term cruise altitude. Use 3 digits for altitude or flight level. Note -- recomputes handoff sectors by doing a track reroute from present position.

Action	Requirements	Command	Notes
Assigned Altitude and Uplink	CPDLC eligibility	QZ /U ddd <FLID ♦> QZ ddd /U <FLID ♦>	Also uplinks the altitude to the pilot, as "climb and maintain" or "descend and maintain". See CPDLC chapter for description of green underline, wilco, etc.
Interim Altitude		QQ ddd <FLID ♦>	Enter temporary altitude during climbs and descents. 1 to 3 digits.
Interim Altitude and Uplink	CPDLC eligibility	QQ /U ddd <FLID ♦> QQ ddd /U <FLID ♦>	Also uplinks the altitude to the pilot, as "climb and maintain" or "descend and maintain". See CPDLC chapter for description of green underline, wilco, etc.
Delete Interim Altitude		QQ <FLID ♦>	Remove temp altitude.
Interim Altitude	Data block not owned.	QQ /TT ddd <FLID ♦> QQ /TT <FLID ♦>	1 to 3 digits. Don't know if in ERAM.
Reported Altitude		QR ddd <FLID ♦> QR <FLID ♦> QR /OK ...	Second option reports level at the assigned or interim altitude. If Mode C is being received, the datablock will not show the QR altitude. If Mode C is not received (non-radar), the most recent QR altitude will appear in the datablock.
Uplink Cross Fix at Altitude, optional Speed	CPDLC eligibility.	UC <Fix> A<alt> <FLID ♦> UC <Fix> A<alt> S<speed> <FLID ♦>	Must use a Fix named in the flight plan. No abbreviated names. General limits as in popup dialog – Altitude (FL) 010-600, Speed 100-380.
A.I.T.	Data block handing from a Center sector to your Center sector.	<FLID ♦> QQ ddd <FLID ♦> <Sector ID> <FLID ♦>	Approximates an Automated Information Transfer (AIT) procedure. Accept handoff, enter interim (optional), initiate handoff, each within 8 seconds. Previous sector assigns the altitude you enter and switches comm to the following sector. See General chapter in eATS.PDF.
Track Reroute		QU <fixes/left clicks ...> <FLID or RIGHT click on position symbol> QU LAX N1234X QU SEA PDX N1234X QU /OK ...	<fixes> is a series of fixes or LEFT click(s) on the map. Final fix must be point in current flight plan. (Type it, or click tolerance is 1.5 miles.) Present position automatically added before first point. Displays route line for 10 seconds. Can't change destination yet. Airplane needs to be told the new route. Use AM if airways involved.
Track Reroute and Uplink (one fix)	CPDLC eligibility	QU /U <fix> <FLID/right click> QU <fix> /U <FLID/right click>	Also uplinks "direct to <fix>" to the pilot. See CPDLC chapter for description of green underline, wilco, etc. Use full fix name (1 or 2 characters is a radio option).

Action	Requirements	Command	Notes
Enter 4 th Line Heading /Speed		QS ddd <FLID ♦> QS /ddd <FLID ♦> QS ddd/ddd <FLID ♦> QS /OK ...	Enter heading and/or /speed. Heading is ddd. Speed is /ddd. May use /Mdd or /M.dd for mach number. May append '+' or '-' to speed or mach. Replaces 4 th line data.
Enter 4 th Line /Speed and Uplink	CPDLC eligibility	QS /U /ddd <FLID ♦> QS /ddd /U <FLID ♦> QS /U /RNS <FLID ♦> QS /RNS /U <FLID ♦> QS /OK ...	Also uplinks the speed, mach, or Resume Normal Speed to the pilot. See CPDLC chapter for description of green underline, wilco, etc. RNS is displayed after the uplink and before a response is received.
4 th Line Text		QS =<text> <FLID ♦> QS /OK ...	"=" seems like a good symbol for Windows keyboards. Replaces 4 th line data.
Delete 4 th Line Data		QS * <FLID ♦> QS */ <FLID ♦> QS /* <FLID ♦> QS /OK ...	Delete all 4 th line data, or just heading, or just speed. If all deleted, uses Data Block menu selection for line 4 Destination.
Show/Hide 4 th line data		QS <FLID ♦> Click on HSF symbol	Heading-Speed-Free Form indicator (⇄) is shown to the right of datablock line 3 if corresponding line 4 QS data is present. If QS data is hidden or blank, uses Data Block menu selection for line 4 Destination.
VCI on/off		//<FLID ♦> Click at VCI position	Turn Voice Communication Indicator on/off. Symbol location and click area is just left of ERAM datablock line 2. VCI must be on for CPDLC dialog to show the frequency change option. D-side updates automatically if allowed by item on A.I. menu. May change your entry.
Release CPDLC eligibility		RE <FLID ♦>	Details in CPDLC chapter, Datalink Status section.
Steal CPDLC eligibility	VCI On. Current eligibilty is within the active Center, or not assigned.	SX <FLID ♦>	Details in CPDLC chapter, Datalink Status section.
Code Request		QB <AID/CID/♦>	Computer assigns a discrete transponder code, e.g. oceanic to domestic.
Code Modification		QB oooo <AID/CID/♦>	Assign a specific code, 4 octal digits. Rejected if discrete code in use by another aircraft.

Action	Requirements	Command	Notes
Enter Flight Plan	Flight plans are normally read from the data files. Since this function is probably used for testing, you can use it without being plugged in.	FP Call Sign Type TAS Fix [D E]XX00 FL Route Airborne: N1234X C172 105 LAX EXX00 80 SAN../LAX..SFO On ground: DAL123 A359 480 KLAX DXX00 340 KLAX..RZS..BSR..KSFO	Only time option so far is XX00 (= now). Flight will become active in a sweep or two. Use time EXX00 to start airborne. Fix must match item after "../." in route. 'E' refers to estimated time. XX00 means offset zero minutes from now. Use time DXX00 to start on the ground. Fix must match first item in route. 'D' refers to departure time.
Show Flight Plan		QF <FLID ♦>	Output format is: CID AID Type Beacon TAS Fix Time Altitude ReqAlt Route Remarks. For now route elements have spaces between to allow the edit window to automatically wrap to the next line.
Route Readout		AM <FLID ♦>	Show ATC flight plan route (field 10). This option has the proper dots instead of spaces, but will not wrap to the next line.
Beacon		AM <FLID> BCN oooo AM <FLID> 04 oooo AM <FLID> 4 oooo	Change beacon code. BCN is field 4 (of 11) in Show Flight Plan/QF description above. Similar to QB/Code Modification.
Assigned Altitude		AM <FLID> ALT (d)dd AM <FLID> 08 (d)dd AM <FLID> 8 (d)dd	Change assigned altitude. ALT is field 8 (of 11) in Show Flight Plan/QF description above. Similar to QZ/Assigned Altitude.
Requested Altitude		AM <FLID> RAL (d)dd AM <FLID> 09 (d)dd AM <FLID> 9 (d)dd	Change pilot requested altitude. Work in progress to make consistent with 3 other cruise altitudes. RAL is field 9 (of 11) in Show Flight Plan/QF description above.
Route		AM <FLID> RTE route AM <FLID> 10 route	See Field 10 Route Amendments section below.
AM accepts multiple field pairs. Delete field contents not programmed. AM N1234X BCN 4531 ALT 80 RAL 110			
----- DISPLAY ----- (Most do not require plugged in)			
Data Block Offset		<direction> <FLID ♦> / <length> <FLID ♦> <direction> / <length> <FLID ♦>	Direction is 1-9 oriented as computer number pad (1-2-3 on bottom). Direction 5 sets relative to current track. Length is 0 to 5. Search eATS.PDF for "1-2-3" for instructions if you have a 1-2-3 on top number pad.

Action	Requirements	Command	Notes
Force Data Block	Data block not displayed	<FLID> or Left-click on first target symbol (backslash).	Display data block.
Force Data Block	Data block displayed but not owned.	<FLID ♦>	Remove (not owned) data block.
Request Data Block	Data block not displayed. Target tracked.	QP <FLID> or QP Left-click on first target symbol (backslash).	Display data block. Does not "accidentally" accept handoff if it starts flashing at you at the same time.
Suppress Data Block not implemented	Data block displayed.	QP <FLID ♦>	Not implemented yet.
Point Out	Plugged in.	QP <Sector ID> <FLID ♦>	Display data block at another sector, which doesn't do anything, but you can make the entry. May be useful if an interphone system is programmed. Sector ID is a 2 digit internal sector, or <letter><number><number> for an external Center sector. More error conditions to detect.
Point Out Approved	Plugged in. Data block not owned.	QP A <FLID ♦> Click on P above datablock	If P shows above non-owned data block, approves the point out. AI hands to next sector at appropriate time.
Remove Point Out Indicator	Plugged in. Data block owned.	QP A <FLID ♦> Click on P above datablock	If P shows above owned data block, removes the P. You did not accept the point out before accepting the handoff. Airplane was/will be transferred to your frequency. To simplify the programming, if you accept the handoff and leave the P displayed, it will be removed when the next sector accepts your handoff.
5 mile Halo	Target displayed. Does not have to be owned by player.	QP J <FLID ♦>	If no halo or a 3 mile halo, displays a 5 mile halo. If 5 mile halo, turns halo off.
3 mile Halo	3 mile menu item selected. Airplane at or below FL230. ERAM datablock format. Target displayed. Does not have to be owned by player.	QP T <FLID ♦> QP J 3 <FLID ♦>	If no halo or a 5 mile halo, displays a 3 mile halo. If 3 mile halo, turns halo off. A 3 mile halo becomes a 5 mile halo if an airplane climbs above FL230.
Reposition Strips (In Sector and Outbound)		QP S Left-click	Click defines lower left corner of list. List grows upward. Intended for lower portion of screen.

Action	Requirements	Command	Notes
Reposition Inbound List		QP I Left-click ^----- letter	Click defines upper left corner of list. List grows downward. Intended for upper portion of screen.
Reposition Departure List		QP D Left-click	Click defines upper left corner of list. List was mostly for debugging time calculations for departure spacing, and is now initialized off-screen. Click somewhere on-screen if you want to see it.
Reposition Metering List		QP M Left-click	Click defines upper left corner of list. List grows downward. Intended for upper portion of screen.
Reposition Keypad Stack		QP K Left-click	The Zoom buttons, CRD, QAK keys, Number Pad, and Comm receive/transmit windows may be moved as a group to a new location. The click position defines the new lower right corner, adjusted to avoid off-screen left.
Route Display		QU <FLID> QU RIGHT click on position symbol	Display/remove route line for FLID. Up to 3 may be shown. All disappear 10 seconds after last entry. Can ignore timeout using QU/Track menu item. Other menu items show climb/descent calculation points and the pilot route. Start point of track is only updated with a Track Reroute. The orange dot in the "show all data" option is the first step in making the start point move appropriately. QU left-click makes the entry a Track Reroute, described in the Flight Data section.
Route Display Off		QU <enter>	Removes all QU route lines now.
Quick Look		QL < sector/airport ID(s) > QL 36 43 QL LAX QL <enter> QL ALL	Shows full data blocks in selected (non-active) sectors within facility, and/or airplanes with matching destination airports. Adds to existing list up to 5 items. Turn off with no ID. ALL to show everything.

Action	Requirements	Command	Notes
Altitude Limits		QD dddBddd	QD 078B242 displays all limited data blocks between 7800 feet and FL242. Normally the limits are 1200 feet above and below your sector's altitudes (to make traffic calls – not implemented yet). Many functions that change the sector configuration automatically adjust the limits, and will change the values that you set. Limited data blocks for all aircraft that will be handed off to you are shown regardless of the QD limits.
Altimeter		QD <airport ID> AR <airport ID>	Display one altimeter setting. May use 3 or 4 letter ID. Station must be named in a *WX entry in the facility DATA file. (Commands similar to real system, but program does not display in the same manner.)
Conflict Suppress/Request		CO <FLID ♦> <FLID ♦> Mouse click(s) processed on present position slash of a full data block, but if none are nearby will use a limited data block.	Stop (or re-enable) blinking of a pair of data blocks. Conflict logic not processed every sweep, might take 15 seconds. Changing active sectors clears all suppressed pairs. Useful when not plugged in and you can't do anything about it, or for traffic in adjacent sectors not separated by the A.I.
Strip Request		SR <FLID ♦>	On/off. Force 2 line format for In Sector strip when within radar coverage. Non-radar In Sector always uses 2 line. Outbound strips are always 1 line. Note – 2 line strip marking is tied to radio communication. Minimal updates when not plugged in.
Upper Winds		UR	One set of winds at a time for everywhere. Table shows flight level, true direction, speed. Winds change on selecting a facility, and with WIND and NEWWINDS commands below.

Action	Requirements	Command	Notes
----- CONTINUOUS RANGE READOUT -----			
Define Group Name		LF //<fix> LF //<fix> <Group> LF //WYNDE LF //WYNDE W LF //WYNDE090010 W10	If <fix> is 5 characters or less, Group Name does not have to be typed, and will be the same as the <fix>. Fixes with more than 5 characters (e.g. lat/longs) must specify Group Name. Group Name may be 1 to 6 characters. "ALL" is not allowed, because it has special meaning in CRR entries below. The same fix may be used in multiple Groups. All LF // definitions are deleted when a new facility is loaded. User Data files may be used to store LF entries for each sector. See Data Files PDF, search for LF.
Show/Hide Range Readout for Airplane(s)		LF <Group> <FLID ♦> LF <Group> FLID/FLID/... LF W <click> LF WYNDE 182 LF W10 234/765/N1234X	Mouse click OK if one airplane. Multiple airplanes (no limit) may be separated with '/'. If an airplane had no Group assignment or was assigned to a different Group, the range to the fix associated with the specified Group is displayed. If an airplane was already assigned to the specified Group, the airplane is removed from the Group and the range readout disappears. Expected use is with CIDs, but other forms of Flight ID also work.
Items with similar functions are only controlled by mouse actions in the real system			
List		CRR LIST	Lists current LF // definitions. May not fit in the CRD output window. Note – some database errors will prevent CRRs from loading. Check Log File.

Action	Requirements	Command	Notes
Set Characteristic of one Group		CRR <Group> <Value>	<Value> may be: DELETE deletes LF // definition SMALL use small datablock font LARGE use large datablock font WHITE or W set color GREEN or G set color PINK or P set color YELLOW or Y set color
Set Characteristic for All Groups		CRR ALL <Value>	If <value> is a font or color, changes the display of all current ranges, and sets default for new LF // Groups for this run. Program is initialized with SMALL and WHITE. ALL DELETE (or loading a different facility) deletes all LF // definitions.

Command	Example	Action
----- FACILITY AND SECTOR -----		
XX FAC LLL	XX FAC ZMP XX FAC ZMP SEC 16	Make facility active. Erases all airplanes. May select a sector in the same command. Not plugged in after change. Produces new altimeter settings and surface wind. Centers defined – ZAN, ZHN, ZSE, ZOA, ZLA, ZLC, ZAB, ZMP, ZDV, ZKC, ZFW, ZHU, ZAU, ZID, ZME, ZOB, ZTL, ZMA, ZJX, ZDC, ZNY, ZBW.
XX SEC dd	XX SEC 16	Make sector active. Not plugged in after change. Think walking across the control room to another sector.
XX COMBINE dd XX COMB dd	XX COMBINE 15 XX COMB 15	Add another active sector. Keeps existing airplanes. Will stay plugged in. May enter as one command: XX FAC ZOA SEC 29 COMB 32
XX REMOVE dd	XX REMOVE 16	Makes sector inactive. Keeps other active sectors, keeps existing airplanes, will stay plugged in. Possibly less confusing term than "split".
XX MID dd XX MID <area>	XX MID 32 XX MID A	If midnight shift configuration is defined in the data files, selects/combines/activates all sectors within a work area. Configs are displayed in XX LIBRARY window. Does not change clock time or time-of-day traffic selection.
XX CLOSE Fac Sec XX OPEN Fac Sec	XX CLOSE ZAU 23 XX OPEN MSN 1A	Changes FPA ownership based on *CLOSE target in sector definition. Used to split Super High airspace for assignment to multiple sectors below, and for taking over Tracon airspace after hours (5 mile separation). Need both Fac and Sec in this command. Other sector selection operations only apply to sectors within the active facility, and Fac is not included in those commands. May take some time to build new ATC Tracks for each airplane. Fast if used before airplanes are active.
XX CLOSE ALL XX OPEN ALL		Opens or Closes all Sectors and Tracons that have *CLOSE targets defined in the database. CLOSE ALL is likely to be appropriate for the midnight shift. Normally CLOSE on an active sector is rejected. CLOSE ALL is accepted if there is more than one active sector, which means it can be used after XX MID without extra steps. When a different facility is loaded, all sectors are initially open. When the same facility is reloaded, closed sectors remain closed.
XX CLOSE LIST		Shows which Sectors and Tracons are able to be closed (have *CLOSE targets). A list of currently closed sectors is always displayed in the upper left.
XX RELOAD		Erases all airplanes and reloads same facility and active sectors. One use is to re-read data files after editing. Will stay plugged in. No confirmation dialog, but after all that typing I assume you meant to do it.

Command	Example	Action
XX WORK		Use instead of mouse clicks on menu items and check boxes. As necessary changes status to airplanes on, plugged in, D-side. Typical use on program start up: XX FAC ZBW SEC 22 WORK
XX WATCH		Use instead of mouse clicks on menu items and check boxes. As necessary changes status to airplanes on, unplugged. Typical use on program start up: XX FAC ZBW SEC 22 WATCH
----- TRAFFIC -----		
XX ACGEN		Turns aircraft generator on/off. Existing airplanes continue. The "Airplanes Off" switch on the File menu erases all airplanes right now. Current traffic level percentage remains the same. If was zero, remains zero. If off and any sectors selected, status message in green in upper left.
XX ACGEN ddd	XX ACGEN 75	Sets traffic level from 0 to 200%. Existing airplanes continue, including those approaching the active sectors. May take some time to stabilize at the requested value. The design load is 15-20 airplanes, depending on the type of sector. Traffic is adjusted based on the flight plans defined to operate during the current or selected time of day. Adding a second sector automatically reduces the new traffic generated, because this is expected to be a fairly common operation. If you combine 3 or more sectors during busy times, you will probably want ACGEN at something less than 100%, maybe 65%, and 4 sectors 55% etc. Value is shown in upper left if not 100%. Shown in white (i.e. a normal operation) except 0% shown in green.
XX DAYTRFC		Switches On or Off. Regardless of other time settings, picks from all "daytime" flight plans that operate between 0500 and 2259 facility local time. Displays DAYTIME TRAFFIC near clock. Will generate the most traffic. The only way to see flight plans with times between 2300 and 0459 local is when the local time is within that range (real time or LOCAL time option), and DAYTRFC is turned off.
XX LOCAL dd	XX LOCAL 8 XX LOCAL 23 XX LOCAL -1	Sets local hour for facility in use and adjusts GMT. CLEARS FLIGHTS AND RESTARTS because existing times will be invalid. Use -1 to turn off override and use real time. Uses flight plans and traffic levels for designated facility local time. Displays SIMULATOR TIME near clock. Minutes remain real minutes in case you have an external deadline to be aware of.

Command	Example	Action
XX WARMUP		Resets initial start distance outside the active sector to program default. Corresponds approximately to WARMUP 25, but has more random variation.
XX WARMUP <value>	XX WARMUP 50 XX WARMUP -50	Sets approximate distance from sector boundary where first airplane is created. Large value means it will take longer for a sector to reach the design traffic level (warm up). Value 0 to 100. The first airplane is created on or outside the active sector boundary. The next several airplanes are created at increasing distances, up to a program defined value. Value -100 to -1. Initial airplanes may be created a negative distance outside (i.e. inside) the active sector boundary. Status message Start Within Sector is displayed. More in Enroute chapter in eATS.PDF.
XX DEONLY		Departures only. Switches On or Off. Uses flight plans defined in the Library that start on the ground (not "ORIG./."). A flight may be repositioned airborne if the origin is a long distance from the active sector, but the general effect is to allow working with mostly departures. Turned off if ARRONLY is turned on. Also on Traffic menu.
XX ARRONLY		Arrivals only. Switches On or Off. Uses flight plans defined in the Library that end on the ground (not "/.DEST"). A flight may start on the ground or fly through the active sector if it lands somewhere else, but the general effect is to produce a higher percentage of arrivals. Turned off if DEONLY is turned on. Also on Traffic menu.
XX METERING		Display/hide list if metering items defined for facility/sector. At present does not affect traffic flow. Also on Traffic menu.
XX METER_DETAILS		Display/hide more information for debugging. Also on Traffic menu.
XX FREEZE		Display current value.
XX FREEZE <dd> MIN XX FREEZE <dd> NM	XX FREEZE 30 MIN XX FREEZE 200 NM	If METER_RWY used in facility, set Freeze Horizon in minutes. Minutes may be 30 to 60. Initial value is 30 minutes. If METER_APT used in facility, set Freeze Horizon in miles. Initial value is different in each facility.
XX AAR		Display list of Airport Arrival Rates. If METER_RWY used in facility, lists by landing runway. If METER_APT used in facility, lists by airport.

Command	Example	Action
XX AAR <rw> <dd>	XX AAR KSFO28L 15 XX AAR KSFO 30	If METER_RWY used in facility, change rate for one database metered runway. If METER_APT used in facility, change rate for one airport. Does not accept a new airport definition. Minimum value is 4 airplanes per hour. Either _RWY or _APT may be defined in the database for a facility, but not both at the same time. Definitions can be added to your User Data files, but must match the type used in the distribution files, if any.
----- MAP -----		
XX CENTER <fix>	XX CENTER EGLL	Moves map center to designated point. Can not enter <fix> using a mouse click, but see SHIFT+HOME.
XX RANGE <range>	XX RANGE 75	Set map range. Original command before spin buttons and keyboard options were programmed.
XX ZOOM <fix>	XX ZOOM EAU	Set map center to <fix>, range to 40 miles and turns on fix labels. ZOOM buttons for each sector can be defined in the data files with a radius other than 40. Original command before zoom buttons were programmed.
XX ZOOM NONE		Restores the database center and range. See table below for information on zoom buttons.
XX RECENTER		Restores database center and range. Was original command before ZOOM NONE was programmed
XX GRID		Turns the latitude/longitude grid on/off. Green lines. 1 degree spacing for small ranges and 5 degree spacing for large ranges. Also on a Display menu item.
XX RINGS		Turn range rings (range arcs) on/off. Also on Display menu, Other Lines. Definitions are in Map section of DATA file, or in User Data.
XX LAMB		Sets Lambert projection. Default, and same as used on many aeronautical charts.
XX ORTHO		Sets Orthographic projection. Ortho, grid on, range 3200 looks like a globe. Not needed for this program but the math was there from a previous one.
XX MAGNORTH		Aligns map vertical center line with magnetic north. I have not seen a Center display that uses this format. Code was already in the map logic so I enabled it. Sets projection to Lambert. Selecting a facility and possibly other map actions will reset to true north.
XX TRUENORTH		Aligns map vertical center line with true north. Default.
XX TERRAIN		If terrain data base installed, turns display on/off. GTOPO30, 30 arc-second data with reduced vertical resolution. Also a Display menu switch for separate coastline display from NOAA GSHHG data.

Command	Example	Action
XX TERRAIN <d>	XX TERRAIN 2	Changes terrain display colors. 1 = dim colors, data blocks easier to read (default). 2 = approximate VFR chart colors, data blocks hard to read. 3 = dim grays, better for higher average terrain. 4 = bright grays, better for lower average terrain.
XX BLUE <ddd> XX BLUE		Changes blue background level, or displays current level. Default is 25. Range 0 – 255. Recommend values well below 100.
XX WHITE		Changes map background to white. Might be useful for taking a screen shot, editing, and printing a map, should anyone still be using their printer. Colors do not work well for data blocks and some other elements. Use XX BLUE <ddd> to get back to normal.
XX SHOWLL		On/Off. Displays lat/longs of boundary points.
XX SOLID1		Turns on/off lines defined as SOLID_LINE_1. Also on a Display Menu item. Used to draw special purpose lines as needed. Line type can be turned on by default by a database designation for each sector. Player may then turn on/off as desired.
XX SOLID2		Turns on/off lines defined as SOLID_LINE_2.
XX SOLID3		Turns on/off lines defined as SOLID_LINE_3.
XX AWSEG		Turns on/off lines defined as Airway Segments.
----- RADAR -----		
XX RADARHOLES		Activates/Deactivates holes in radar coverage. Airplane is non-radar if below the surface all radar holes at the current position. Also on Radar menu.
XX SHOWRADARS		Brown. Displays (no-terrain interference) radar horizon of some radar sites at selected MSL altitudes. Used to help adjust radar hole parameters. Not used in calculations. Also on Radar menu.
XX SHOWHOLES		Cyan. Displays location of radar hole surface at selected MSL altitudes. Outer displayed ring is edge of radar hole. Also on Radar menu.
XX RADAR <altitude>	XX RADAR 10000	Gives distance from sea level to geometric (no atmosphere), visual (light refraction), and radar horizons.
----- ENVIRONMENT -----		
XX WINDSON XX WINDSOFF		Turns upper winds on/off. UR command displays winds.
XX NEWWINDS		Creates a new set of winds. Automatic on selecting a new facility and a few other cases. UR command displays winds.

Command	Example	Action
XX WINDdddsss	XX WIND270050	Mostly for testing. Sets winds aloft for <i>all altitudes</i> to given direction and speed. No range checks. Ridiculous values may produce ridiculous results.
XX SFCWIND dir spd	XX SFCWIND 300 10	Use after selecting a facility if you want different values. Direction is rounded to nearest 10 degrees. Speed may be 0 to 30 knots. Value displayed at upper left. The next XX FAC command generates a new surface wind based on the wind distribution defined for that facility.
XX QNH dddd	XX QNH 2973	Set desired altimeter 28.93 to 30.91. Work in progress. Some affected operations respond, others not programmed. Altimeter setting shown at upper left. Random value produced on each XX FAC command. So far should be set before airplanes are created – e.g. after this use XX RELOAD or select Airplanes Off, then On.
XX NEWPRESSFIELD		Set a new random Pressure Field for Individual Station Altimeters, in case you didn't like the previous one. So far should be set before airplanes are created – e.g. after this use XX RELOAD or select Airplanes Off, then On.
----- POP UP WINDOWS -----		
For all pop up windows - If you click on the window, such as to reposition it, Windows directs keyboard and mouse input to that window (not useful). Click on the main window to resume normal inputs. Some function keys will re-select the main window too.		
XX LIBRARY XX LIB XX LIB OFF F10 key		Displays arrival restrictions for active sector and other data in a pop up window. F10 key shows/hides this window and the Debug Tool window.
XX STRIPWIN XX STRIPWIN OFF Ctrl+F10 key Display menu item		Show/hide separate Strip Window. Can be dragged to a second monitor. Also hides/shows normal Inbound and In-Sector strip lists.
(Debug Tool) XX FMC <FLID ♦> XX FMC OFF F10 key	XX FMC N1234X	Displays aircraft flight data in a mixed Airbus/Boeing/made-up Flight Management Computer format. This is a debugging tool, not an airplane simulator. If you know how any FMC operates you will get the idea. Displayed values are NOT always what you would see in a real airplane. Not necessary to use the program. F10 key shows/hides this window and the Library window.
XX FMC XX TRACK XX TRACK <FLID ♦>		Sets Debug Tool window to airplane FMC format or ATC Track values (altitudes, sectors, etc.). Track format can be displayed initially instead of FMC format. Not necessary to use the program.
XX CONFLICT XX CONFLICT OFF		Displays conflict alert data in a pop up window. A.I. does not apply separation except when airplanes are initialized. This is the first step in researching how that might be improved.
XX DATAFOLDER		Same as menu item File, Open Data and Log Folder.

Command	Example	Action
XX KEYPADS Shift+F10 key Display menu item		Show/hide on-screen keypads. Moves comm windows and Zoom buttons. Moves In Sector strip list if database position is "Right", and the player has not moved the list with QP S.
----- SIMULATION CONTROL AND INFORMATION -----		
XX PAUSE		Stops or starts the program clock. When paused, the comm window, other XX functions, and mouse clicks within the map area are disabled. After restarting, and until the next database re-read, "TIME OFFSET" is displayed to show that program minutes and seconds are not in sync with your computer clock. Some items that only affect the display continue to operate. Results may not be apparent until un-paused. A function that would normally use a mouse click, such as repositioning a data block, will work if you type the CID or call sign.
XX INFO <identifier>	XX INFO KLAS XX INFO V13 XX INFO KLAX-ANJLL4 XX INFO KEGE-LDA25	Shows data base information about airports, nav aids, airways, SIDs, STARs, approaches. There is a hyphen between an airport and a procedure name. Zero substitution on procedure names not programmed. Procedures with multiple transitions and long airways will probably overflow the CRD display area. Transitions and breaks in airways are separated with "=".
XX LOG <FLID ♦>		FMC and Track values to log file for one aircraft. Similar to data in Debug Tool window. Contents change depending on what I need for debugging. File menu option opens folder containing LogFile.txt.
XX ALERT		Turn on/off alert sound played when an airplane initiates most conversations. Also on File menu. Start Option available.
XX COMMQ		Print airplane pending communications queue to log file.
XX COMMQCLEAR		Clear pending communications queue for everybody. Can use if an airplane won't respond to acknowledgements and quit calling you. A less severe alternative is transmitting "<call sign> ??", meaning "how do you read", which should clear that airplane's comm queue.
XX RPTQ		Print airplane pending report queue to log file – report leaving, report entering hold, etc.
XX FREQ <Fac> <Sec>	XX FREQ ZAB 68	Frequencies you need are almost always displayed in the Strips List. Data for external sectors is only loaded for sectors adjacent to the active facility. This function does not handle single letter facility IDs.
XX DEST3ON XX DEST3OFF		Turns off/on single letter data block destination code in line three (if defined in facility data files). Also on a Data Block menu item.
XX DBCOLOR		Changes data block color between yellow and white.

Command	Example	Action
XX FONTS		Switches on/off. Displays a list of fixed pitch fonts available on your system in 8 point. Last line is END OF LIST if all fit on your screen. Log file shows fonts requested by the program on start up. Up arrow, down arrow, and caret are from the Unicode characters found in most fonts. User control of font/size will be added to the program sometime. Method to be determined.
XX ERAMFONTS		Switches on/off. Displays a group of weights and sizes for one or two fonts selected by an *ERAM_TEST_FONTS entry in eATS.txt. More in Data-Files PDF, User Data chapter.
XX AUTORUN		Executes a set of XX commands defined within an AUTORUN block in eATS.txt. Example in Data-Files.pdf
----- DATABASE CONSISTENCY -----		
XX MINUTERUN	XX MINUTERUN ZSE ZAB ZLC	Not useful unless you are doing extensive database modifications. Runs each facility for about a minute. Check the log file for errors.
XX CHECKFPLIB		Not useful unless you are doing database modifications. Checks the flight plan library for the active facility for errors, and writes to log file. The only checks so far are for flight plans that start airborne or end airborne within the active facility. MINUTERUN also does this for each facility.
XX LOGENTERSFAC <Fac ID>	XX FAC ZAU WATCH XX LOGENTERSFAC MKE	Writes flight plans to the Log File that enter the requested facility from the active facility. Example extracts active ZAU flight plans that go through MKE airspace. Copy results from Log File to User defined Tracon flight plan file. May take 5 minutes to process every flight plan.
----- CALCULATOR AND MORE -----		
XX DRAW <space separated elements> XX DRAW <field 10 format with '.'> XX DRAW XX REDRAW	XX DRAW OBH 45N090W GCO180020 KLAX25L XX DRAW SFO TRUKN0 DEDHD RBL ILS34 RDD XX DRAW SFO.TRUKN0.DEDHD. .RBL.ILS34.RDD	Draws solid line between known elements. Draws dotted line when skipping unknown elements to highlight errors. If SID, STAR, or approach is used, also include origin/destination airport. Handles airport alias and procedure "zero" designations. Can type XX DRAW<space> and then control+V (paste) text copied from an external source. XX DRAW removes line. REDRAW reinstates previous DRAW, and leaves command in input area for further editing, or to recompute wind-aware procedures after changing the surface wind. <ESC> will clear input area.

Command	Example	Action
XX MACH <TAS> <FL> [2nd FL] XX MACH <MACH> <FL>	XX MACH 458 350 XX MACH 458 350 390 XX MACH .85 350	If first value is ≥ 1 converts TAS and FL to MACH. If a second FL is given, shows TAS to get computed MACH at the second FL. Assumes standard temperatures. If first value < 1 converts MACH and FL to TAS and IAS.
XX GS-TAS <GS> <FL> <FIX1> <FIX2> <WD> <WS> (optional)<TEMP>	XX GS-TAS 408 360 HEMLO 42N130W 270 85 -55 produces: TAS 484 IAS 281 M 84 TC 243 TH 248 ISA +1	Inverse wind triangle, ground speed to TAS. Calc TAS from observed ground speed, flight level, course, forecast wind. Fixes define true course. Optional temperature (with sign) at altitude from winds aloft forecast.
XX RF <Center> <Fix1> <Fix2>	XX RF N3826.27/W10649.88 WADDU METVY	Compute points to approximate a Radius to Fix RNAV approach leg. See Data-Files PDF.
XX ICAO <route>	XX ICAO DCT DIPSO G595 INVUS/M084F310 G595 ATNAT/M084F330 G595 SOTKI DCT RIGMI DCT 30S170E/M083F350 DCT 27S180E DCT TERAN R582 NUGLI/M084F370 R582 URKEP DCT 21S160W DCT 19S157W DCT 14S150W DCT 06S140W DCT 03N130W/M084F390 DCT 11N120W DCT 20N110W DCT MZT/N0481F390 DCT ALRES DCT PNG DCT SAT	Converts ICAO format route to NAS format removing DCT, speeds, etc. CRD input is a standard window that accepts clipboard data. May copy/paste from somewhere. Results appear in a pop up window, and are written to the log file. Route expected to start with ORIG DCT FIX, or ORIG AIRWAY FIX, or DCT FIX. Route expected to end with FIX or DCT DEST. Example (YSSY-KIAH) produces: ..DIPSO.G595.INVUS.G595.ATNAT.G595.SOTKI..RIGMI..30S170E..27S180E..TERAN.R582.NUGLI.R582.URKEP..21S160W..19S157W..14S150W..06S140W..03N130W..11N120W..20N110W..MZT..ALRES..PNG..SAT

Quick Action Keys

Text Command	Quick Action Key	Function Key	Short Summary	
QA	AUTO HAND		Enable/Inhibit automatic handoff initiation from the active sector for one airplane. Sector/Facility disable not programmed.	
QB	CODE		Request/Modify transponder code.	
QD	CRD		Altitude limits for display of limited data blocks. Shown in upper left corner of screen. Limits were shown on the R-CRD in NAS Stage A.	
QF	SHOW FP	F2	Show ATC flight plan.	
QH	HOLD		Not yet programmed.	
QL	QUICK LOOK		Sector ID(s) separated by spaces, blank to clear, ALL	
QN			"None" key refers to commands that don't require a QAK or even the "QN" prefix, and include accepting and initiating a handoff and positioning a data block.	
QP	PVD	F7	Historical name of the "radar screen" is Plan View Display. Refers to displaying something on a PVD somewhere.	
QQ	INT ALT	F8	Interim altitude. Used during climbs and descents for a temporary altitude. Displays "xxxTxxx" in the data block.	
QR	REPORT		Enter pilot reported altitude (if Mode C not available).	
QS	HDG SPD	F12	4 th line data.	
QT	TRACK		Not yet programmed. Everybody auto-acquires so far in a manner that is convenient for the program.	
QU	ROUTE	F3	Enter flight plan reroute points or display/remove track line. Additional display options on QU/Track menu.	
QX	CANCEL		Drop Track after an airplane has landed. Other uses, but not until start track QT is programmed.	
QZ	ASGN ALT	F5	Used to change a long term cruise altitude. Uses '^', 'v', and 'C' in the data block. Affects target handoff sectors. QZ vs. QQ had other implications in NAS Stage A regarding printing strips.	
CO	CO		Conflict Suppress/Request (blinking data blocks)	
FP	ENTER FP		Create a new flight plan. Details above.	
AM	AM		Flight plan amendment.	
DM	DM		Not yet programmed. Departure Message. Function is automatic in a manner that is convenient for the program.	
	RNG BRG	F4	Compute distance and bearing between points. Other options described above.	
	HOME		Cursor to center of map area.	

Keyboard Shortcuts

Windows Keyboard KEY	FUNCTION
HOME	Move cursor to map center. Not the physical screen center because of the buttons, etc. on the right side.
SHIFT + HOME	Move map center to cursor.
PAGE UP	Increase vector line length.
PAGE DOWN	Decrease vector line length.
SHIFT + PAGE UP	Increase map range.
SHIFT + PAGE DOWN	Decrease map range.
F1	
Function key F2	"QF " Show flight plan
Function key F3	"QU " Route display/modify
Function key F4	Range-bearing
Function key F5	"QZ " Assigned Altitude
F6	
Function key F7	"QP " PVD
Function key F8	"QQ " Interim Altitude
F9	
Function key F10	Library and FMC windows on/off
Shift + F10	On screen Keypads on/off
Ctrl + F10	Strip Window on/off.
F11	
Function key F12	"QS " 4 th Line Heading/Speed/Text
ESC or ESCAPE	Erases contents of CRD or Transmit window without taking any action.
SHIFT + hover over present position slash	Show flight plan and route line (QF QU). May be replaced by a more realistic function if available. Looks at full data blocks first, but if none are nearby will use a limited data block.

Zoom Buttons

click on ZOOM NONE button

Restores the center and range in effect before other Zoom buttons were used. This will be a manually set center and range, or the database center and range if no manual adjustments were made.

Changing the sector configuration sets the database center and range. Subsequent adjustments using spin buttons, XX commands, menu items, or keystrokes will be remembered when Zoom None is used (until the next sector configuration change).

Shift + click on ZOOM NONE button

Restores database center and range. Same response as menu item Display, Zoom None.

The programming is designed for two styles.

If you like the database center/range and only use zoom buttons, ZOOM NONE and Shift+ZOOM NONE do the same thing.

If you manually adjust the center/range after loading a sector, ZOOM NONE gets you back to the last manual center/range, and Shift+ZOOM NONE sets the database values.

click on other zoom buttons

Sets center and range defined in the DATA file. Also displays fix labels.

If you want to remove the labels while zoomed, use menu item Display, Symbols, Labels or the Labels check box. It may take 2 clicks to get the display in sync with the check marks.

CAUTION – if the mouse is positioned over a zoom button, some Windows functions are inhibited, probably because the active window is considered to be the button. Clicking on any zoom button automatically moves the cursor off the button to avoid this problem.

Field 10 Route Amendments

If a route amendment contains only fixes, either AM or QU may be used. The most common route change is expected to be present position direct to a fix in the current flight plan, and that QU will be used. The options available for field 10 correspond to the radio commands available for route changes.

Airway/SID/STAR/Approach modifications must use AM. Field 10 may be specified with "10" or "RTE". The program has limited use for field 6 (fix) and field 7 (time), and no user-modifications to those fields are programmed.

The first AM route element is a fix (any of the legal formats) near the present position of the airplane. You do not have to be exact. The updated track will begin at the position you enter. The present position is not part of the radio command. The D-side has an uncanny ability to eyeball the position of the airplane for the initial AM fix.

The last fix of the new route must be in the current route. If the current route ends with FIX./DEST, only fixes up to FIX are considered. If the last fix of the new route is not the destination airport, the previous route after the fix is retained. If the last fix is the destination airport, and the destination airport does not appear twice in the route, all of the original route is replaced. Changing the destination airport is not programmed.

To simplify internal bookkeeping, STARs end at the destination airport. Most Stars end within a Tracon, so there is little need to have both a Star and an instrument approach in the same route in this simulation. Note – you may add a STAR or IAP to a route that did not have one originally, or change an existing STAR or IAP to a different one, or remove either by clearing direct to the airport.

Both a radio clearance and a field 10 update may end with a STAR or IAP. The destination is automatically added.

Examples, current flight plan contains RBL and destination is RDD:

```
AM N1234X RTE FMG130025..SWR..RBL
AM N1234X 10 FMG130025..SWR.V494.SAC..RBL
AM N1234X RTE FMG130025..RBL.ILS34.RDD
AM N1234X RTE FMG130025..RBL.ILS34
```

Examples for the discussion in eATS.PDF, Departures, Changing the Route.

Initial direct to fix:

```
AM EJA1 RTE DNW.V465.TOCUD..JEKUG
```

Different SID, CID is 698:

```
AM 698 10 ALPIN3.KICNE..JEKUG
```

After surface wind change:

```
AM DAL1 10 RV1.UPP
```